

(0.20) (a) In cylindrical coordinates, Eq. (10.31) becomes

$$\begin{aligned} \frac{\vec{E}^* \cdot \vec{A}_{sc}}{D_0} &= \frac{k^2}{4\pi} \frac{\delta\epsilon}{\epsilon} (\vec{E}^* \cdot \vec{E}_0) \int e^{i\vec{q} \cdot \vec{x}} d^3x = \frac{k^2}{4\pi} \frac{\delta\epsilon}{\epsilon} (\vec{E}^* \cdot \vec{E}_0) \int_{-L}^L e^{iq_{\parallel}z} dz \int_0^{2\pi} d\theta \int_0^a \rho d\rho e^{i q_{\perp} \rho \cos\theta} \\ &= \frac{k^2}{4\pi} \frac{\delta\epsilon}{\epsilon} (\vec{E}^* \cdot \vec{E}_0) \frac{e^{iq_{\parallel}L/2} - e^{-iq_{\parallel}L/2}}{iq_{\parallel}} \cdot \int_0^a 2\pi \rho J_0(q_{\perp}\rho) d\rho \\ &= \frac{k^2}{4\pi} \frac{\delta\epsilon}{\epsilon} (\vec{E}^* \cdot \vec{E}_0) \frac{2 \sin(q_{\parallel}L/2)}{q_{\parallel}} \cdot 2\pi \frac{q_{\perp} a J_1(q_{\perp}a)}{q_{\perp}^2} \\ &= \frac{k^2 a^2 L}{4} \frac{\delta\epsilon}{\epsilon} (\vec{E}^* \cdot \vec{E}_0) \frac{\sin(q_{\parallel}L/2)}{q_{\parallel}L/2} \cdot \frac{2 J_1(q_{\perp}a)}{q_{\perp}a} \end{aligned}$$

Averaging over polarization, the differential cross section is

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= \frac{k^4 a^4 L^2}{16} \left| \frac{\delta\epsilon}{\epsilon} \right|^2 \frac{1}{2} (1 + \cos^2\theta) \left| \frac{2 J_1(q_{\perp}a)}{q_{\perp}a} \frac{\sin(q_{\parallel}L/2)}{q_{\parallel}L/2} \right|^2 \\ &= \left| \frac{\delta\epsilon}{\epsilon} \right|^2 \frac{k^4 a^4 L^2}{32} (1 + \cos^2\theta) \left| \frac{2 J_1(q_{\perp}a)}{q_{\perp}a} \frac{\sin(q_{\parallel}L/2)}{q_{\parallel}L/2} \right|^2 \end{aligned}$$

(b) For $ka \ll 1$, $q_{\perp}a \ll 1$, we have $2J_1(q_{\perp}a)/q_{\perp}a \approx 1$.

The average reduces to

$$\left\langle \frac{d\sigma}{d\Omega} \right\rangle = \left| \frac{\delta\epsilon}{\epsilon} \right|^2 \frac{k^4 a^4 L^2}{32} (1 + \cos^2\theta) = \frac{1}{q} \int_0^q \left(\frac{\sin(q_{\parallel}L/2)}{q_{\parallel}L/2} \right)^2 dq_{\parallel}$$

By integration by parts,

$$\begin{aligned} \frac{1}{q} \int_0^q \left(\frac{\sin(q_{\parallel}L/2)}{q_{\parallel}L/2} \right)^2 dq_{\parallel} &= \frac{1}{qL/2} \int_0^{qL/2} \left(\frac{\sin x}{x} \right)^2 dx \\ &= \frac{1}{qL/2} \left(-\frac{\sin^2 x}{x} \Big|_0^{qL/2} + \int_0^{qL/2} \frac{\sin 2x}{x} dx \right) \\ &= \frac{2}{qL} \int_0^{qL} \frac{\sin x}{x} dx - \left(\frac{\sin(qL/2)}{qL/2} \right)^2 = \frac{2}{qL} Si(qL) - \left(\frac{\sin(qL/2)}{qL/2} \right)^2 \end{aligned}$$